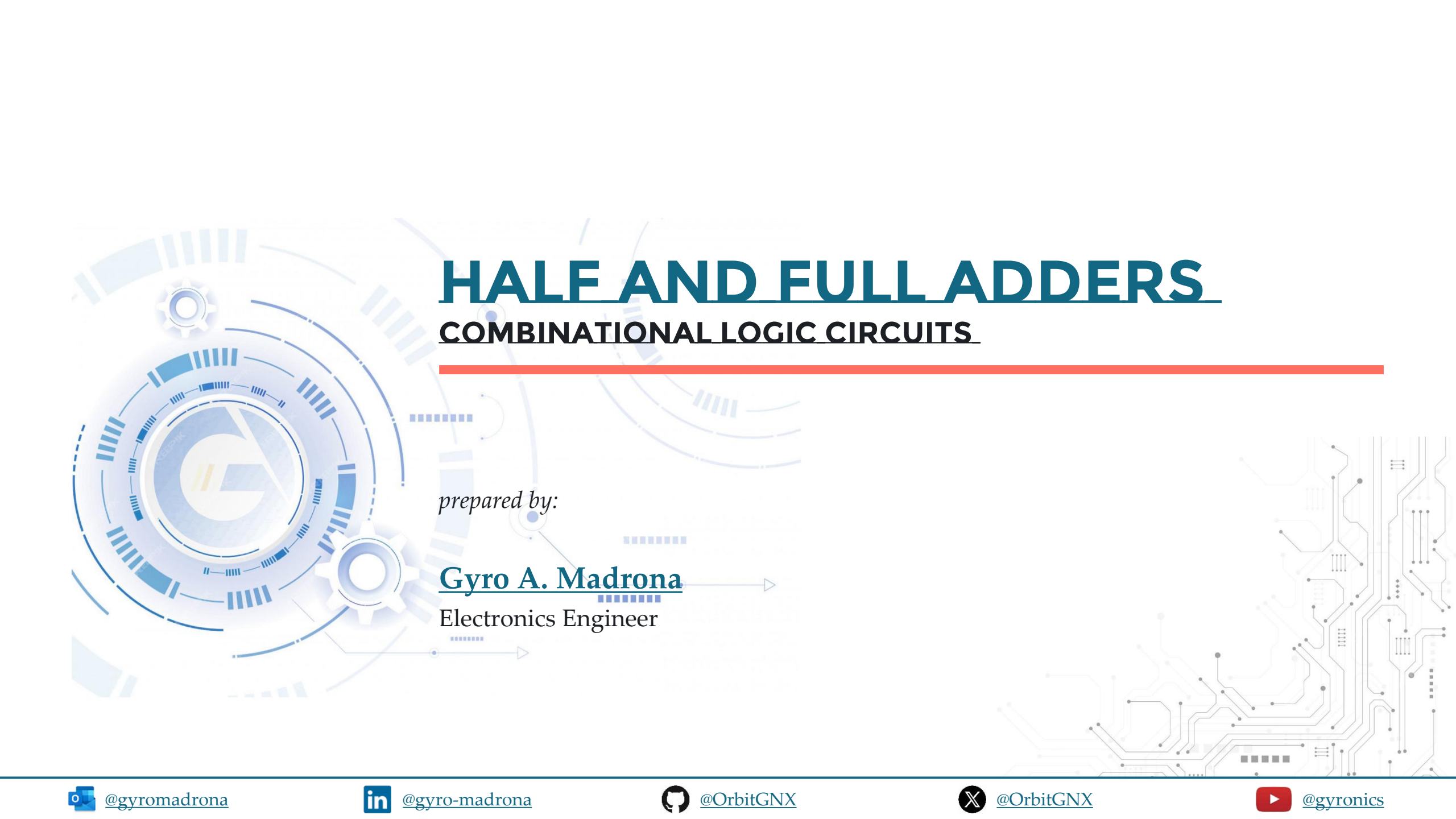




HALF AND FULL ADDERS

COMBINATIONAL LOGIC CIRCUITS

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Half-Adder

Full-Adder



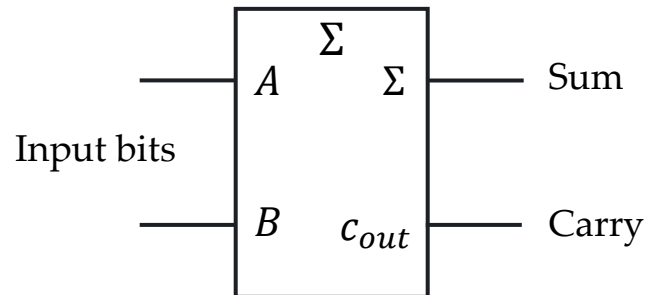
HALF-ADDER



HALF-ADDER

The half-adder accepts two binary digits on its inputs and produces two binary digits on its outputs – a sum bit and a carry bit.

Logic Symbol



Truth Table

A	B	C_{out}	Σ
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0



EXERCISE

Using the truth table of a half-adder, implement the minimized expressions for both the Sum and Carry outputs.

Solution

Truth Table

A	B	C_{out}	Σ
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0



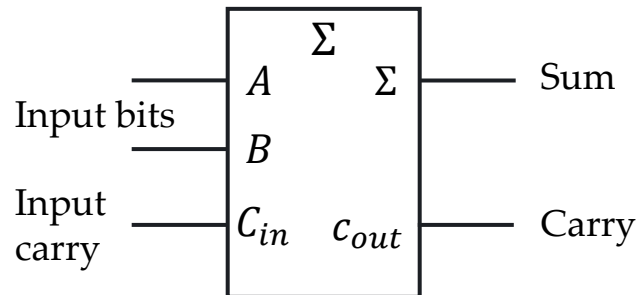
FULL-ADDER



FULL-ADDER

The full-adder accepts two input bits and an input carry and generates a sum output and an output carry.

Logic Symbol



Truth Table

A	B	C_{in}	C_{out}	Σ
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

EXERCISE

Using the truth table of a full-adder, implement the minimized expressions for both the Sum and Carry outputs.

Solution

A	B	C_{in}	C_{out}	Σ
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



EXERCISE

Create a block-level representation of a 2-bit binary adder using full-adder modules.

Solution



EXERCISE

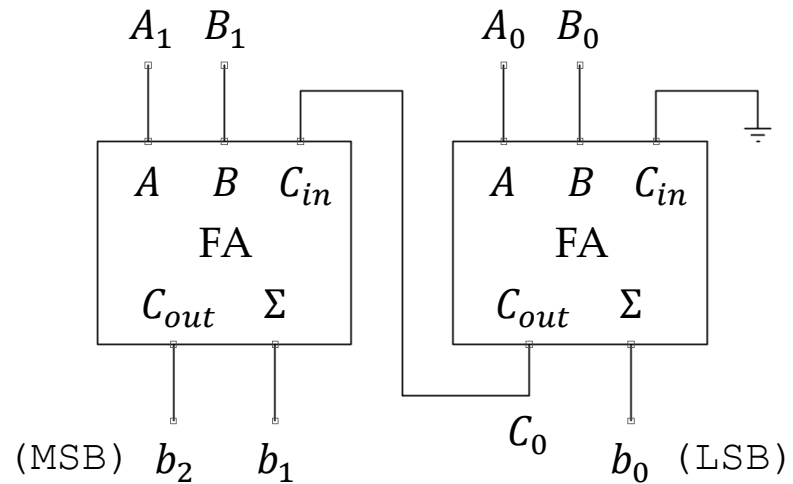
Create a block-level representation of a 3-bit binary adder using full-adder modules.

Solution



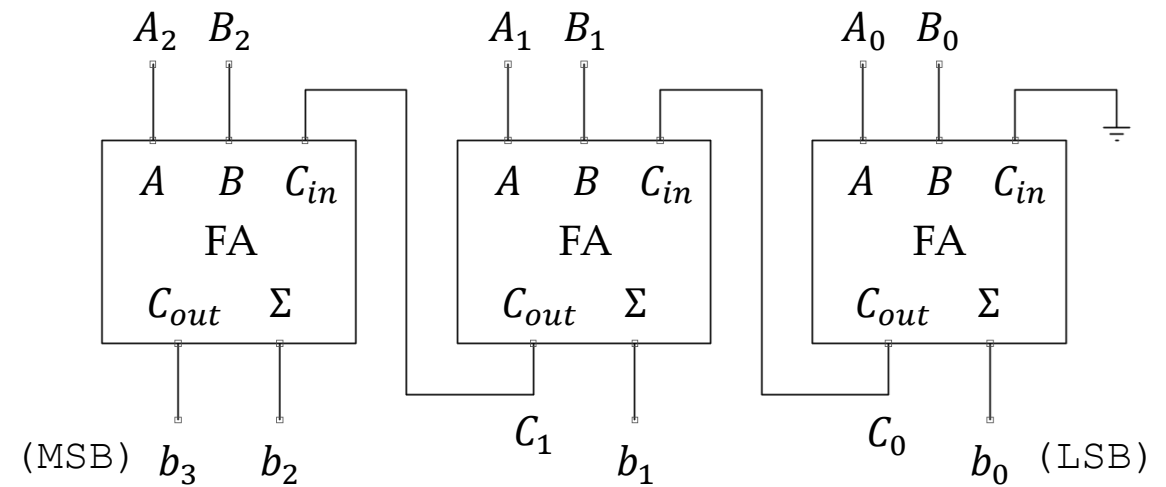
PARALLEL BINARY ADDERS

2-bit Parallel Adder



$$\begin{array}{r} C_1 \ C_0 \\ A_1 \ A_0 \\ B_1 \ B_0 \\ \hline b_2 \ b_1 \ b_0 \end{array}$$

3-bit Parallel Adder



$$\begin{array}{r} C_2 \ C_1 \ C_0 \\ A_2 \ A_1 \ A_0 \\ B_2 \ B_1 \ B_0 \\ \hline b_3 \ b_2 \ b_1 \ b_0 \end{array}$$



EXERCISE

Implement a 2-bit parallel binary adder without utilizing XOR or XNOR gates.

Solution



LABORATORY

